

# Amazon SDE Intern(6M) Experience

**About the company:** Amazon, one of the MAANG companies, is a global leader in e-commerce, cloud computing, and AI. Known for its innovation and customer obsession, Amazon empowers engineers to build scalable, high-impact solutions that serve millions of users worldwide.

**Role Offered:** Software Development Engineer Intern

**Stipend:** 1,10,000 INR + 425 USD (Relocation Allowance) per month

**Duration:** 6 Months

**Selection Process:**

1. Online Assessment (Virtual)
2. Technical Interview (Virtual/In-Person)

Amazon came for 2 roles, one is for SDE intern role another is for Non-SDE role intern. For Non-SDE intern they shortlisted the candidates by CGPA around 8.75.

For SDE intern role, the online assessment happened on Oct 3, 2025 on Virtual Mode. The assessment contains 3 Parts

1. Coding round (2-DSA) - Timed for 70 mins
2. Workstyle Assessment - No Time restriction
3. Work Simulation Assessment - No Time restriction

There are a plenty of ways in which you can get into amazon. They conduct variety of events like *Hackon, ML Challenge:* These 2 programs are Hackathons via this programs you can get opportunities to enter into amazon and can win prizes.

*ML Summer School:* 1-Month Virtual Learning program via this you get into ML Scientist or related roles.

Usually these programs happen in between May- September, So Keep an eye on LinkedIn, unstop and Tech Influencers, They'll notify you about it. I participated in all these programs.

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## **OFF-CAMPUS EXPERIENCE:**

I got a Invitation for SDE Intern role through my performance in Amazon Hackon in July 2025. And then, I cleared the OA then got an interview link before the campus placements started(Aug). But unfortunately I Couldn't clear it. I was really nervous since my last attended interview was in 5th sem Nov 2024 through On-campus.

The interviewer began by introducing himself and asked my introduction and my project for around 10 mins in Amazon Chime Platform then I was asked to code and explain 2 DSA qns although I correctly coded the soln, I couldn't explain it to the interviewer qns why I'm using BFS instead DFS then why 3D visited array instead of a 2D visited Array for 1st qn.

Then the 2nd qn was a famous modified binary search question, I asked for testcases then asked for range of the variable in that question, then I gave the binary search soln, He asked me what was the brute force approach I gave a approach using backtracking by producing continuous subset sum within the threshold and the no of subsets is the no of days. I haven't seen the problem before, he was looking a for loop approach which was super-intuitive instead of while loop(Binary search) he was expecting that, But i have a different, he couldn't understand my backtracking approach i guess.

Then he asked if I have any question I asked 2-3 qns then the interview ended, within a week a got a mail that I was not selected. *Key learning after this incident was to improve pattern-based learning for DSA although I knew the optimized approach, I have to start with a brute-force soln and discuss it's time and space complexity then I have to move the optimized soln and code that soln out.*

Online Assessment: [OA Question Link](#)

### ONLINE ASSESSMENT:

Then days pass by on Oct 3, I wrote amazon OA through on-campus this time the DSA questions were tough. One of the question was on DP another was a Simulation based question.

Question 1 – Dataset Cleaning Cost You are given a string dataset of even length consisting of lowercase English letters. You can remove any two characters in one operation:

If both are the same → cost = matchCost

If different → cost = mismatchCost

Your goal is to remove all characters while minimizing the total cost.

Testcase 1:

dataset = "ouio", matchCost = 2, mismatchCost = 4

Here the matchCost is lower than mismatchcost so need to match as many chars as possible

'o' - 'o' got matched cost=2 'u','i' left in the string

'u' - 'i' got mismatched so cost=cost+mismatchcost=2+4=6 string is now empty so return 6

Testcase 2:

s = "ouio", matchCost = 6, mismatchCost = 4

Here the matchcost > mismatchCost so try to mismatch chars as possible to reduce the cost

'o'-'u' cost=4

'i'-'o' cost=cost+mismatch=4+4=8 so return 8

Testcase 3:

s='ooouio' matchcost=6 mismatchcost=4

Here we have to match a 'o'-'o' no other way so ans would 14

From the question I figured out we are trying out different possibilities and choosing the minCost, So figured it as a Dp problem and coded a soln that passed 13/15 Testcases.

Question2 - You have a pipeline string and a failedService character.

Each second, every occurrence of failedService removes its preceding character (if any).

All removals happen simultaneously.

Continue until no more removals possible.

Output = number of seconds until stabilization.

TestCase 1:

pipeline = "database", failedService = 'a'

At second 1:'a' preceding chars are removed after that pipeline="aaase"

At second 2:'a' preceding chars are removed pipeline="ase"

now 'a' got no preceding so the string got stabilized so return the no seconds which is 2

I coded a  $O(n*m)$  soln which eventually got time limit exceeded for the last testcase.

n - size of the string

m - no of seconds

I try to log the input that failed using print statements and the input was a very large string, I tried to optimise it but couldn't.

### WORKSTYLE ASSESSMENT:

They gave situation where I was new joinee in the company and i was asked a write documentation for codebase and the manager was not satisfied with my documentation. How would reply to various scenarios of question. Try to put yourself there and select the options that defines responsibility ,ownership and curiosity.

### WORK SIMULATION ASSESSMENT:

Be consistent same type of question will be twisted in many ways you have to select the option which is more likely your value and be consistent throughout.

For Example:

Completing a project within timeline or Enjoying Freetime

You have to select 1st one: *Completing a project within timeline*

Then after some question: *Do you like wear a watch for office or I like to read books*

Now select the 1st one *Do you like wear a watch for office*

Both the questions implies that you value time which one of the principles of amazon.

Select the option that more talk about team ownership, frugality, amazon's core leadership principles. Try to read about it online before the assessment.

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After a couple of days, the shortlist came and nearly 100+ student were shortlisted and after a week nearly 18 students were selected. I was not shortlisted for interview.

Yes, I was not shortlisted. But I already had experience of amazon interview and know about the pattern what they focus more. So helped my friend who were shortlisted for interview rounds to focus more on Binary search/Modified Binary search Problems and graph problems especially the problems related to shortest path like Rotten orange and Connected components problems and gave a list of almost must do 30 problems.

And exactly on the interview he was asked a problem on connected components and Koko eating banana problems he solved both the problem and after the interview he said both the problems were from the list of 30 problems which is said. And he was selected.

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Then days passed, I was applying for other companies and hopeless. On Oct 27, Amazon again shortlisted 100+ candidates for SDE intern role and the interview was scheduled on Nov 7 and Nov 8 on Sathyabama engineering College, Chennai and it was an offline interview and I was shortlisted.

Until oct 27, I have solved around 400+ DSA problem on Leetcode out of which 200 problems were done at my third year. So I solved only 200+ problems during July - Oct during placement season.

But within those 10 days prepration time, I solved around 120+ new problems and revised all those patterns and problem I have solved already. Finished the graph, DP, Trie series of Take you Forward Playlist.

### **OFFLINE INTERVIEW:**

On Nov 7, I went to the interview location and seriously I was expecting a Shortest path using Dijkstra Algorithm/BFS, Topological sort, Minimum Spanning Tree, Trie-based string matching or Any Tree Based questions But Surprisingly I was given Buy and Sell Stock problem, the interviewer explained the question like Amazon Stock's daily fluctuating between value and you have to buy and sell the stock and maximise the profit. I asked some clarifying question on the question since there were many variations on Buy & sell stock which could be solved using DP, So I asked whether I could Buy and sell only 1 time or multiple times and any codition for buying stocks since these were other variants. The interview said You could buy and sell only once. So I confirmed that this is the variant one.

```
int profit=0,buy=profit[0];
for(int sell:stocks){
    if(sell-buy>profit){
        profit=sell-buy;
    }else
    {
        buy=min(sell,buy);
    }
}
return profit;
```

I explained the problem using a testcase, either we can buy or sell at point,so we have to buy at the min stock price and sell at the maximum profit.

Then discussed the Time & space complexity: I said Tc:O(n) and Sc:O(1)

then the interviewer asked how it is  $O(1)$  space since i was using 3 variable.  
I explained even if the profit array contains 10 elements or 1 million elements the 3 variables gonna remain the same so it is a constant space.

Time & Space complexity is a measure of how the variables gonna increase regarding as the size of the input. Let's say I'm usign a array it gonna store all the (buy-sell) at every point and choosing the maximum profit iterating the array it gonna take  $O(n)$  space.

Both the interviwers were happy with my explanation and asked the 2nd question.

The 2nd question was a LinkedList question were I was asked Reorder the List.

I gave away 3 different approaches:

1. Creating a new list with the same copy of value and reversing it, then merging one from the given list and another from the reversed list.

```
class Solution {
public:
    void reorderList(ListNode* head) { //I Coded the soln in paper
        ListNode* temp=head;
        ListNode* revList=new ListNode(0);
        ListNode* temp1=revList;
        int n=0;
        while(temp){
            n++;
            temp1->next=new ListNode(temp->val);
            temp1=temp1->next;
            temp=temp->next;
        }
        revList=reverse(revList->next);
        ListNode* ans=new ListNode(0);
        temp=head;temp1=ans;
        for(int i=0;i<n;i++){
            if(i>=ceil((double)n/2)){
                break;
            }
            temp1->next=temp;
            temp=temp->next;
            temp1=temp1->next;
            if((n & 1 && i!=n/2) || (n%2==0)){
                temp1->next=revList;
                revList=revList->next;
                temp1=temp1->next;
            }
        }
        temp1->next=NULL;
        head=ans->next;
    }

    ListNode* reverse(ListNode* p){
        ListNode *q=nullptr,*r;
        while(p){
            r=q;
            q=p;
            p=p->next;
            q->next=r;
        }
    }
};
```

```

    }
    return q;
}
};

```

Approach 2: Instead of creating a new list I gave an approach using stack which will store the list in reverse order which is also of time complexity of  $O(n)$  and space complexity of  $O(n)$ .

Approach 3: I gave an In-place technique where I use constant space.

Using slow-fast pointer technique I will find the middle of the list and reverse the 2nd part alone.

Then picking one from the start and another from the reversed list and merging it.

TC: $O(n)$  SC: $O(1)$

After this approach, the interviewer seemed very happy and I took almost 30-40 mins to solve both the question. then, I gave a copy of my resume and finished the interview.

1 week later after completing my 1st semester exam in KP, My frnd said that amazon shortlist came and there was no internet connection in the 2nd floor, We rushed to the ground floor and saw the list, there were 25 candidates selected and to my surprise my name was not there.

Then after completion the semester exams, After Dec 5, the placements resumed, Companies registrations started, I got shortlisted for *Media.net* 2nd Round for Site reliability Engineer role.

On Dec 12, I was in Media.Net socket programming coding round, My frnd came in the middle of test and laughed and came to me and said "Come on, leave the test". And then he showed the message that I was offered 6month Internship for Amazon SDE Role. It happened in the most dramatic way Ever!

***FINALLY, THE YOUNG SEAGULL MADE IT'S FIRST FLIGHT!***

### KEY EXPERIENCE & LEARNINGS:

- Research about the company's recent OA and hiring pattern, what they're looking at. And what topics they're focusing more on.  
For Eg : Amazon - DSA, Media.net - Computer networks + OS + System Design ,  
Visa - Project Understanding(System Design) + Famous DSA problems,  
Nvidia - Low-level DSA like bit manipulation, pointers, C/Cpp
- Certain interviews will be different from the others, Media. net - 2nd round will be a Socket programming where we have to implement client-server programs like group chat rooms, accessing folders, multiple client + one server handling. 4th round will be system design round  
Visa - In the virtual coding rounds they ask us to implement code from the scratch, when I was attending the interview for intern(2M) they asked to implement a height and depth of a tree, where I have to include the header files, tree creation, Linkedlist creation etc... (Most of the candidates struggle here)
- Search experience in forums like Reddit, LeetCode discuss, GFG, Medium & Youtube etc... In Nvidia, They have been asking the same qns in OA for almost 3 years, I saw it in a leetcode discuss post but didn't practise the qn but for my surprise the next day I got the same qns.
- For DSA, Complete the [Striver's A-Z sheet](#) Completely since 75% of interview question will be in that sheet, Some other Imp questions that are not in striver's sheet will be in [NeetCode](#), for Best soln & famous problems - [Aryan Mittal](#)
- Computer networks - [IT K Funde Channel](#)
- OS qns related to Memory management - [BitLemon](#)
- Follow some famous influencers on LinkedIn, Instagram & Youtube who will be posting about off-campus opportunities, hackathons.
- Participate in coding/tech events in Kurukshetra, Sync, Itrix and also in Hackathons, You end up learning new things or winning it which will be significantly become turning point in Interview and also open up new avenues like internships opportunities etc...

- Try to store the dsa soln in digital form preferably Github, google sheets/Locally which will help us to revise pattern wise - Constantly Revise concepts and pattern wise problems.
- Don't worry about outcomes in life we are not playing a T20 match, we are playing a test match just hang in there is always a second innings.

**Must Visit CS Fundamentals & DSA preparation resources that I used:** [My resources](#)

*If you want something you've never had, you have to do something you've never done!*

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#### **CONTACT:**

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*Everything will be fine in the end, If it is not then it is not the End!*